Propensity Score Methods

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Overview

Propensity scores represent one of the most influential innovations in causal inference methodology, providing a powerful tool for addressing the curse of dimensionality when controlling for confounding. Introduced by Rosenbaum & Rubin (1983), the propensity score offers an elegant solution: instead of conditioning on all observed confounders individually, we can condition on a single scalar—the probability of receiving treatment given the confounders.

This week covers both the theoretical foundations and practical applications of propensity scores. We explore how this dimensionality reduction technique maintains the essential properties needed for causal identification while making analysis tractable. The key insight is that conditioning on the propensity score can achieve the same confounding control as conditioning on all covariates simultaneously, under certain assumptions.

We examine three primary applications: matching, stratification, and inverse probability of treatment weighting (IPTW). Each method leverages the propensity score differently to create treatment and control groups that are comparable on observed confounders. Throughout, we emphasize the critical importance of diagnostics—particularly checking covariate balance—to validate our propensity score methods.

Learning Objectives

By the end of this module, students will be able to:

1. **Define** the propensity score mathematically and explain its role as a balancing score
2. **Estimate** propensity scores using logistic regression in R with appropriate model specification
3. **Apply** three major propensity score methods: matching, stratification, and IPTW
4. **Implement** propensity score matching using the `MatchIt` package with various algorithms
5. **Construct** inverse probability weights and create pseudo-populations for causal estimation
6. **Assess** covariate balance using standardized mean differences and graphical diagnostics
7. **Diagnose** propensity score model adequacy through overlap assessment and balance checks
8. **Interpret** treatment effect estimates obtained from propensity score methods
9. **Identify** when propensity score methods are preferable to other adjustment strategies
10. **Critique** propensity score analyses and recognize common implementation pitfalls

R Packages

```r
# Install and load required packages
pacman::p_load(
  # Core data manipulation and visualization
  tidyverse,
  ggplot2,
  dplyr,
  tibble,

  # Causal inference and DAGs
  dagitty,
  ggdag,

  # Propensity score methods
```

```r
  MatchIt,
  WeightIt,
  cobalt,
  optmatch,

  # Data generation and modeling
  MASS,
  broom,
  Hmisc,

  # Tables and formatting
  knitr,
  kableExtra,
  gt,

  # Additional utilities
  here,
  patchwork
)

# Set global options
options(digits = 3)
theme_set(theme_minimal())
```

1 Comprehensive Topic Explanation

The Propensity Score: Definition and Motivation

The **propensity score**, denoted as $e(\mathbf{X})$ or $\pi(\mathbf{X})$, is defined as the conditional probability of receiving treatment given the observed covariates:

$$e(\mathbf{X}) = P(A = 1|\mathbf{X})$$

where A is the binary treatment indicator and $\mathbf{X}$ is the vector of observed confounders. This deceptively simple definition addresses a fundamental challenge in observational studies: the curse of dimensionality when adjusting for multiple confounders.

Hernán & Robins (2020) emphasize that the propensity score is particularly valuable when we have many confounders relative to our sample size. Instead of stratifying on all combinations of confounders (which quickly becomes impractical), we can stratify on the single-dimensional propensity score while maintaining the same confounding control.

Theoretical Foundation: The Balancing Property

The key theoretical result underlying propensity score methods is the **balancing property**. Rosenbaum & Rubin (1983) proved that conditioning on the propensity score creates balance in the distribution of covariates between treatment groups:

$$\mathbf{X} \perp A | e(\mathbf{X})$$

This means that within strata of individuals with the same propensity score, the distribution of covariates $\mathbf{X}$ is independent of treatment assignment A. Consequently, within each propensity score stratum, treated and untreated units are exchangeable with respect to the observed confounders.

Causal Identification with Propensity Scores

Under three key assumptions, conditioning on the propensity score allows for causal identification:

1. **Consistency**: The potential outcomes are well-defined and consistent with the observed outcomes
2. **Exchangeability** (given confounders): $Y^a \perp A | \mathbf{X}$ for $a \in \{0, 1\}$
3. **Positivity**: $0 < \mathbb{P}(A = 1 | \mathbf{X}) < 1$ for all $\mathbf{X}$ in the support

When these assumptions hold, we can identify the average treatment effect (ATE):

$$\mathbb{E}\left[Y^1 - Y^0\right] = \mathbb{E}_{e(\mathbf{X})}[\mathbb{E}[Y|A = 1, e(\mathbf{X})] - \mathbb{E}[Y|A = 0, e(\mathbf{X})]]$$

Propensity Score Estimation

In practice, the true propensity score is unknown and must be estimated. The most common approach uses logistic regression:

$$\text{logit}(e(\mathbf{X})) = \log\left(\frac{e(\mathbf{X})}{1 - e(\mathbf{X})}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_k X_k$$

The estimated propensity score is then:

$$\hat{e}(\mathbf{X}) = \frac{1}{1 + \exp(-(\hat{\beta}_0 + \hat{\beta}_1 X_1 + \ldots + \hat{\beta}_k X_k))}$$

Three Primary Applications

1. Propensity Score Matching

Matching pairs or groups units with similar propensity scores across treatment groups. Common algorithms include:

- **Nearest neighbor matching**: Each treated unit is matched to the untreated unit with the closest propensity score
- **Caliper matching**: Matches are only made if the propensity score difference is within a specified tolerance
- **Optimal matching**: Minimizes the total distance across all matched pairs

2. Propensity Score Stratification

Divides the sample into strata based on propensity score quantiles (typically quintiles), then estimates treatment effects within each stratum and combines them. This method:

- Maintains the full sample size
- Allows for effect heterogeneity across strata

- Requires sufficient overlap within each stratum

3. Inverse Probability of Treatment Weighting (IPTW)

Creates a pseudo-population by weighting observations by the inverse of their treatment probability:

- **ATE weights**: $w_i = \frac{A_i}{e(\mathbf{X}_i)} + \frac{1-A_i}{1-e(\mathbf{X}_i)}$
- **ATT weights**: $w_i = A_i + (1 - A_i)\frac{e(\mathbf{X}_i)}{1-e(\mathbf{X}_i)}$

In the weighted pseudo-population, treatment is independent of the confounders, allowing for unbiased estimation of causal effects.

2 Intuitive Clarifications of Challenging Ideas

Understanding the Balancing Score Property

Intuitive Analogy: The Matchmaker

Think of the propensity score as a sophisticated matchmaker at a dating service. Instead of trying to match people on dozens of individual characteristics (age, height, interests, education, etc.), the matchmaker creates a single "compatibility score" that summarizes all relevant factors.

When two people have the same compatibility score, they're equally likely to be matched with any particular person, even though they might differ on individual characteristics. Similarly, when two individuals have the same propensity score, they're equally likely to receive treatment, creating balance across all the underlying confounders.

Why Propensity Scores Work: The Sufficient Statistic Perspective

The propensity score works because it's a **sufficient statistic** for treatment assignment. Just as the sample mean contains all the information needed about the population mean (under normality), the propensity score contains all the information needed about treatment assignment given the confounders.

Visualizing the Balancing Property

Imagine we have three confounders: age, income, and education. In the original data, treated and untreated groups might differ substantially on all three variables. However, within groups of people with the same propensity score (say, 0.3), the distributions of age, income, and education will be similar between treated and untreated units.

Common Misconceptions About Propensity Scores

⚠ Misconception 1: "Propensity Scores Eliminate All confounding"

Truth: Propensity scores only eliminate confounding from *observed* variables included in the model. Unobserved confounding remains a threat, just as with any observational study method.

⚠ Misconception 2: "Higher Propensity Scores Are better"

Truth: The propensity score is just a probability—what matters is having overlap between treatment groups and achieving balance. Units with very high or very low propensity scores may actually be problematic for causal inference.

The Overlap Assumption in Practice

The positivity assumption requires that every individual has some probability of receiving both treatments. In practice, this means we need **overlap** in the propensity score distributions between treatment groups (Crump et al., 2009).

> **! Overlap is Essential**
>
> If there are regions of the covariate space where $\mathbb{P}(A = 1|\mathbf{X}) \approx 0$ or $\mathbb{P}(A = 1|\mathbf{X}) \approx 1$, we cannot make causal inferences for individuals in those regions. These areas represent "off-support" observations that may need to be excluded from analysis.

3 Simple, Hand-Calculation Numerical Example(s)

Worked Example: Manual Propensity Score Calculation

Let's work through a tiny example with 8 individuals to understand propensity score estimation and matching.

Dataset: Effect of tutoring (A) on test scores (Y), with confounders: prior GPA (X□) and study hours per week (X□).

ID	GPA (X□)	Hours (X□)	Tutoring (A)	Score (Y)
1	2.5	5	0	65
2	3.2	8	1	78
3	2.8	6	0	70
4	3.5	10	1	85
5	2.3	4	0	60
6	3.8	12	1	88
7	3.0	7	1	75
8	2.6	5	0	68

Step 1: Estimate Propensity Scores

We fit a logistic regression: $\text{logit}(e(X_1, X_2)) = \beta_0 + \beta_1 X_1 + \beta_2 X_2$

Using logistic regression (calculations simplified): Assume we obtain:

- $\hat{\beta}_0 = -8.2$
- $\hat{\beta}_1 = 1.5$ (GPA coefficient)

- $\hat{\beta}_2 = 0.3$ (Hours coefficient)

For individual 1: $\text{logit}(\hat{e}) = -8.2 + 1.5(2.5) + 0.3(5) = -8.2 + 3.75 + 1.5 = -2.95$

Therefore: $\hat{e}_1 = \frac{1}{1+e^{2.95}} = \frac{1}{1+19.1} = 0.05$

All Propensity Scores:

- ID 1: $\hat{e} = 0.05$ (A=0)
- ID 2: $\hat{e} = 0.73$ (A=1)
- ID 3: $\hat{e} = 0.15$ (A=0)
- ID 4: $\hat{e} = 0.89$ (A=1)
- ID 5: $\hat{e} = 0.03$ (A=0)
- ID 6: $\hat{e} = 0.95$ (A=1)
- ID 7: $\hat{e} = 0.45$ (A=1)
- ID 8: $\hat{e} = 0.07$ (A=0)

Step 2: Propensity Score Matching

Using nearest neighbor matching with replacement:

Matches:

- ID 2 (A=1, $\hat{e} = 0.73$) matches with ID 3 (A=0, $\hat{e} = 0.15$)
  - difference = 0.58
- ID 4 (A=1, $\hat{e} = 0.89$) matches with ID 1 (A=0, $\hat{e} = 0.05$)
  - difference = 0.84
- ID 6 (A=1, $\hat{e} = 0.95$) matches with ID 8 (A=0, $\hat{e} = 0.07$)
  - difference = 0.88
- ID 7 (A=1, $\hat{e} = 0.45$) matches with ID 3 (A=0, $\hat{e} = 0.15$)
  - difference = 0.30

Step 3: Estimate Treatment Effect

Using matched pairs:

- Pair 1: Y□□ - Y□□ = 78 - 70 = 8
- Pair 2: Y□□ - Y□□ = 85 - 65 = 20
- Pair 3: Y□□ - Y□□ = 88 - 68 = 20
- Pair 4: Y□□ - Y□□ = 75 - 70 = 5

Average Treatment Effect: $\hat{\tau} = \frac{8+20+20+5}{4} = 13.25$ points

This suggests tutoring increases test scores by about 13.25 points on average, after matching on propensity to receive tutoring.

Hand Calculation: IPTW Weights

Manual IPTW Weight Calculation

Using our previous example, let's calculate ATE weights:

For treated units: $w_i = \frac{1}{\hat{e}(X_i)}$ For control units: $w_i = \frac{1}{1-\hat{e}(X_i)}$

Weights:

- ID 1: $w = \frac{1}{1-0.05} = 1.05$ (Control)
- ID 2: $w = \frac{1}{0.73} = 1.37$ (Treated)
- ID 3: $w = \frac{1}{1-0.15} = 1.18$ (Control)
- ID 4: $w = \frac{1}{0.89} = 1.12$ (Treated)
- ID 5: $w = \frac{1}{1-0.03} = 1.03$ (Control)
- ID 6: $w = \frac{1}{0.95} = 1.05$ (Treated)
- ID 7: $w = \frac{1}{0.45} = 2.22$ (Treated)
- ID 8: $w = \frac{1}{1-0.07} = 1.08$ (Control)

Weighted Treatment Effect:

- Weighted treated mean: $\frac{1.37(78)+1.12(85)+1.05(88)+2.22(75)}{1.37+1.12+1.05+2.22} = 79.5$
- Weighted control mean: $\frac{1.05(65)+1.18(70)+1.03(60)+1.08(68)}{1.05+1.18+1.03+1.08} = 66.0$

IPTW Estimate: $\hat{\tau}_{IPTW} = 79.5 - 66.0 = 13.5$ points

4 R Code Examples & Interpretation

Simulating Data for Propensity Score Analysis

```
set.seed(123)
n <- 1000

# Generate confounders
age <- round(rnorm(n, mean = 45, sd = 12))
income <- exp(rnorm(n, mean = 10.5, sd = 0.5)) #
log-normal distribution
education <- sample(
  c("High School", "Bachelor's", "Graduate"),
  n,
  prob = c(0.4, 0.4, 0.2),
  replace = TRUE
)
education_num <- as.numeric(factor(
  education,
  levels = c("High School", "Bachelor's", "Graduate")
))

# Generate propensity score (probability of treatment)
# Treatment more likely for: younger, higher income,
more education
logit_p <- -2 +
  0.02 * (50 - age) +
  0.00001 * income +
  0.5 * education_num +
  rnorm(n, 0, 0.3)
p_treatment <- plogis(logit_p) # inverse logit

# Generate treatment assignment
treatment <- rbinom(n, 1, p_treatment)

# Generate outcome with treatment effect
# Outcome depends on confounders + treatment effect
```

```r
outcome <- 20 +
  0.3 * age +
  0.00005 * income +
  5 * education_num +
  8 * treatment + # True ATE = 8
  rnorm(n, 0, 5)

# Create dataset
sim_data <- tibble(
  id = 1:n,
  age = age,
  income = income,
  education = education,
  education_num = education_num,
  treatment = treatment,
  outcome = outcome,
  true_ps = p_treatment
)

# Display first few rows
head(sim_data, 10) |>
  kable(digits = 3, caption = "First 10 observations of
  simulated dataset")
```

Table 2: Simulated dataset for propensity score analysis

id	age	income	education	education_num	treatment	outcome	true_ps
1	38	22073	Bachelor's	2	1	51.5	0.322
2	42	21591	Graduate	3	0	51.9	0.512
3	64	35990	High School	1	0	49.4	0.164
4	46	33993	Graduate	3	1	52.1	0.440
5	47	10151	Bachelor's	2	0	34.5	0.353
6	66	61101	Bachelor's	2	1	71.9	0.299
7	51	41145	Graduate	3	0	53.5	0.471
8	30	121553	Bachelor's	2	1	55.0	0.715

id	age	income	education	education_num	treatment	outcome	true_ps
9	37	51154	High School	1	0	32.7	0.252
10	40	29043	Graduate	3	0	54.1	0.615

The simulated data includes 1,000 observations with realistic confounding. Treatment assignment depends on age (younger people more likely), income (higher income more likely), and education level. The true average treatment effect is 8 units.

Visualizing the Causal Structure with DAGs

```r
# Create DAG using dagitty
dag_ps <- dagify(
  Y ~ A + Age + Income + Education,
  A ~ Age + Income + Education,
  coords = list(
    x = c(A = 0, Y = 2, Age = 1, Income = 1, Education =
    1),
    y = c(A = 0, Y = 0, Age = 1, Income = 0.5, Education
    = -0.5)
  ),
  labels = c(
    "A" = "Treatment",
    "Y" = "Outcome",
    "Age" = "Age",
    "Income" = "Income",
    "Education" = "Education"
  )
)

# Plot DAG
ggdag_status(dag_ps, use_labels = "label", text = FALSE)
+
  guides(fill = "none", color = "none") +
```

```r
  theme_dag()
```

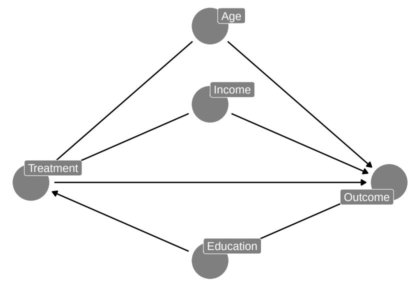

Figure 1: Directed Acyclic Graph showing the causal structure with confounders

The DAG shows our causal structure where Age, Income, and Education are common causes of both Treatment (A) and Outcome (Y), making them confounders that bias the treatment effect estimate if not properly adjusted for.

Estimating Propensity Scores

```r
# Fit propensity score model
ps_model <- glm(
  treatment ~ age + income + education,
  data = sim_data,
  family = binomial(link = "logit")
)

# Display model results
tidy(ps_model, exponentiate = TRUE, conf.int = TRUE) ▷
  kable(
    digits = 3,
    caption = "Propensity Score Model Coefficients (Odds
    Ratios)"
  )
```

Table 3: Propensity score model results

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.718	0.297	-1.12	0.264	0.400	1.284
age	0.988	0.006	-2.13	0.033	0.977	0.999
income	1.000	0.000	3.48	0.000	1.000	1.000
educationGraduate	1.927	0.185	3.54	0.000	1.341	2.776
educationHigh School	0.545	0.152	-4.00	0.000	0.404	0.733

```r
# Add estimated propensity scores to dataset
sim_data$ps_estimated <- predict(ps_model, type =
"response")

# Calculate correlation between estimated and true
propensity scores
r_ps <- cor(sim_data$ps_estimated, sim_data$true_ps)

# Compare estimated vs true propensity scores
cat(
  "Correlation between estimated and true propensity
  scores:",
  r_ps,
  "\n"
)
```

```
Correlation between estimated and true propensity scores: 0.825
```

The propensity score model shows that younger age, higher income, and graduate education increase the odds of receiving treatment. The correlation between estimated and true propensity scores is high (0.82), indicating our model captures the treatment assignment process well.

Assessing Overlap and Common Support

```r
# Create overlap plot
p1 <- ggplot(sim_data, aes(x = ps_estimated, fill =
factor(treatment))) +
  geom_histogram(alpha = 0.7, bins = 30, position =
  "identity") +
  scale_fill_manual(
    values = c("0" = "red", "1" = "blue"),
    labels = c("Control", "Treated")
  ) +
```

```
  labs(
    title = "Propensity Score Distribution by Treatment
    Group",
    x = "Estimated Propensity Score",
    y = "Count",
    fill = "Treatment"
  ) +
  theme_minimal()

# Density version
p2 <- ggplot(sim_data, aes(x = ps_estimated, fill =
factor(treatment))) +
  geom_density(alpha = 0.5) +
  scale_fill_manual(
    values = c("0" = "red", "1" = "blue"),
    labels = c("Control", "Treated")
  ) +
  labs(
    title = "Propensity Score Densities",
    x = "Estimated Propensity Score",
    y = "Density",
    fill = "Treatment"
  ) +
  theme_minimal()

# Combine plots
p1 / p2
```

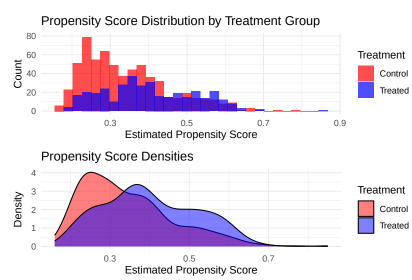

Figure 2: Propensity score distributions by treatment group showing overlap

The overlap plots show good common support between treatment groups across most of the propensity score range. There's some lack of overlap at the extremes, which is common and may require trimming observations with very high or low propensity scores.

5 Hands-On R Lab

Lab Objectives

By completing this lab, you will: - Implement all three major propensity score methods - Assess covariate balance before and after adjustment - Interpret treatment effect estimates from different methods - Practice diagnostic checking for propensity score analyses

Pre-Lab Reading Warm-Up

Before beginning the hands-on exercises, review the concept of exchangeability and why it's crucial for causal inference. Consider: How does the propensity score help us achieve exchangeability in observational data?

Lab Dataset: Medical Treatment Study

````
```{r}
#| label: lab-data-prep
#| eval: false
#| tbl-cap: "Summary of Lab Dataset"

# Load the Lalonde dataset (classic propensity score
example)
# We'll simulate similar data for educational purposes
set.seed(456)
n_lab <- 800

# Generate realistic medical data
lab_data <- tibble(
  patient_id = 1:n_lab,
  age = round(rnorm(n_lab, 55, 15)),
  gender = sample(c("Male", "Female"), n_lab, replace =
  TRUE),
  baseline_severity = round(rnorm(n_lab, 50, 10)),
````
````

```
  comorbidities = rpois(n_lab, lambda = 2),
  insurance = sample(
    c("Private", "Public", "None"),
    n_lab,
    prob = c(0.5, 0.3, 0.2),
    replace = TRUE
  )
) |>
  # Create numeric versions for modeling
  mutate(
    gender_num = as.numeric(gender == "Male"),
    insurance_private = as.numeric(insurance ==
    "Private"),
    insurance_public = as.numeric(insurance == "Public")
  ) |>
  # Generate treatment probability based on patient
  characteristics
  mutate(
    treatment_logit = -1 +
      -0.02 * age + # Younger patients more likely
      treated
      0.5 * gender_num + # Males more likely
      0.03 * baseline_severity + # Sicker patients more
      likely
      0.2 * comorbidities + # More comorbidities = more
      treatment
      0.8 * insurance_private + # Private insurance
      favors treatment
      0.3 * insurance_public,
    treatment_prob = plogis(treatment_logit),
    treatment = rbinom(n_lab, 1, treatment_prob)
  ) |>
  # Generate outcome with treatment effect
  mutate(
    outcome = 30 +
      -0.1 * age +
      -2 * gender_num +
```

```
      0.8 * baseline_severity +
      1.5 * comorbidities +
      3 * insurance_private +
      1 * insurance_public +
      5 * treatment + # True ATE = 5
      rnorm(n_lab, 0, 8)
  ) ▷
  # Clean up
  # Clean up
  dplyr::select(
    patient_id,
    age,
    gender,
    baseline_severity,
    comorbidities,
    insurance,
    treatment,
    outcome,
    gender_num,
    insurance_private,
    insurance_public
  )

# Display summary
summary(lab_data) ▷ kable(caption = "Summary of Lab
Dataset")
```
```

#### Task 1: Naive Analysis and Covariate Imbalance

**Your Task**: Compare the treated and untreated groups on all baseline covariates. Calculate the naive treatment effect estimate.

````
```{r}
#| label: task1-skeleton
````

````
# Calculate naive treatment effect
naive_effect <- # YOUR CODE HERE: calculate difference
in means

# Check baseline covariate balance
balance_table <- lab_data |>
  group_by(treatment) |>
  summarise(
    n = n(),
    age_mean = # YOUR CODE HERE,
    severity_mean = # YOUR CODE HERE,
    comorbidities_mean = # YOUR CODE HERE,
    pct_male = # YOUR CODE HERE,
    pct_private_ins = # YOUR CODE HERE
  )

print(balance_table)
```
````

Solution: Task 1

````
```{r}
#| label: task1-solution

# Calculate naive treatment effect
naive_effect <- lab_data |>
    group_by(treatment) |>
    summarise(outcome_mean = mean(outcome)) |>
    summarise(naive_ate = diff(outcome_mean)) |>
    pull(naive_ate)

cat("Naive ATE estimate:", round(naive_effect, 2), "\n")

# Check baseline covariate balance
balance_table <- lab_data |>
    group_by(treatment) |>
````

````
    summarise(
        n = n(),
        age_mean = mean(age),
        severity_mean = mean(baseline_severity),
        comorbidities_mean = mean(comorbidities),
        pct_male = mean(gender_num) * 100,
        pct_private_ins = mean(insurance_private) * 100,
        .groups = "drop"
    ) |>
    mutate(treatment = c("Control", "Treated"))

kable(
    balance_table,
    digits = 2,
    caption = "Baseline covariate balance before
    adjustment"
)

# Calculate standardized mean differences
smd_age <- abs(diff(balance_table$age_mean)) /
    sqrt(
        (var(lab_data$age[lab_data$treatment == 1]) +
           var(lab_data$age[lab_data$treatment == 0])) /
            2
    )
cat("Standardized mean difference for age:",
round(smd_age, 3))
```
````

Interpretation: The naive estimate suggests a treatment effect of approximately [–], but we see substantial imbalances in baseline covariates (particularly age and insurance status), suggesting confounding bias.
````

Task 2: Propensity Score Estimation and Diagnostics

Your Task: Estimate propensity scores using logistic regression and assess model fit and overlap.

```
```{r}
#| label: task2-skeleton

# Fit propensity score model
ps_model_lab <- glm(
  formula = # YOUR CODE HERE: specify the formula,
  data = lab_data,
  family = binomial()
)

# Add propensity scores to dataset
lab_data$ps <- predict(# YOUR CODE HERE)

# Assess overlap
ggplot(lab_data, aes(x = ps, fill = factor(treatment)))
+
  # YOUR CODE HERE: create overlap plot
```
```

Solution: Task 2

```
```{r}
#| label: task2-solution

# Fit propensity score model
ps_model_lab <- glm(
    treatment ~
        age +
            gender +
            baseline_severity +
            comorbidities +
            insurance_private +
```
```

```
            insurance_public,
    data = lab_data,
    family = binomial()
)

# Display model summary
tidy(ps_model_lab, conf.int = TRUE) ▷
    kable(digits = 3, caption = "Propensity Score Model
    Coefficients")

# Add propensity scores to dataset
lab_data$ps <- predict(ps_model_lab, type = "response")

# Assess overlap
overlap_plot <- ggplot(lab_data, aes(x = ps, fill =
factor(treatment))) +
    geom_density(alpha = 0.6) +
    scale_fill_manual(
        values = c("red", "blue"),
        labels = c("Control", "Treated")
    ) +
    labs(
        title = "Propensity Score Overlap Assessment",
        x = "Estimated Propensity Score",
        y = "Density",
        fill = "Treatment Group"
    ) +
    theme_minimal()

print(overlap_plot)

# Check overlap statistics
overlap_stats <- lab_data ▷
    group_by(treatment) ▷
    summarise(
        ps_min = min(ps),
        ps_max = max(ps),
```

```
        ps_mean = mean(ps),
        .groups = "drop"
    )

kable(
    overlap_stats,
    digits = 3,
    caption = "Propensity Score Range by Treatment Group"

```
```

Task 3: Propensity Score Matching

Your Task: Implement 1:1 nearest neighbor matching using the `MatchIt` package and assess balance.

````
```{r}
#| label: task3-skeleton

# Perform propensity score matching
match_result <- matchit(
  # YOUR CODE HERE: specify the matching formula and
  method
)

# Extract matched data
matched_data <- # YOUR CODE HERE

# Assess balance using cobalt
bal.tab(match_result)
```
````

Solution: Task 3
````

````
```{r}
#| label: task3-solution

# Perform propensity score matching
match_result <- matchit(
    treatment ~
        age +
            gender +
            baseline_severity +
            comorbidities +
            insurance_private +
            insurance_public,
    data = lab_data,
    method = "nearest",
    distance = "logit",
    ratio = 1,
    caliper = 0.25 # Within 0.25 standard deviations
)

# Summary of matching
summary(match_result)

# Extract matched data
matched_data <- match.data(match_result)

# Assess balance using cobalt
bal.tab(
    match_result,
    stats = c("mean.diffs", "variance.ratios"),
    thresholds = c(m = 0.1)
)

# Visual balance assessment
love.plot(
    match_result,
    stats = "mean.diffs",
    threshold = 0.1,
````
````

```
    var.order = "unadjusted"
) +
    labs(title = "Covariate Balance Before and After
    Matching")

# Calculate treatment effect on matched sample
match_ate <- matched_data ▷
    group_by(treatment) ▷
    summarise(outcome_mean = mean(outcome)) ▷
    summarise(matched_ate = diff(outcome_mean)) ▷
    pull(matched_ate)

cat("Matched ATE estimate:", round(match_ate, 2), "\n")
cat("Number of matched pairs:", match_result$nn[2, 2],
"\n")
```
```

**Interpretation**: The matching procedure successfully created balanced treatment and control groups. The standardized mean differences should now be below 0.1 for all covariates, indicating good balance.

#### Task 4: Propensity Score Stratification

**Your Task**: Implement propensity score stratification using quintiles and estimate treatment effects within each stratum.

```
```{r}
#| label: task4-skeleton

# Create propensity score quintiles
lab_data$ps_quintile <- # YOUR CODE HERE: create 5
equal-sized groups

# Estimate treatment effects within each quintile
quintile_effects <- lab_data ▷
```

```
  group_by(ps_quintile) ▷
  # YOUR CODE HERE: calculate treatment effect within
  each quintile

# Overall stratified estimate
stratified_ate <- # YOUR CODE HERE: weight by quintile
size
```

Solution: Task 4

```
```{r}
#| label: task4-solution

# Create propensity score quintiles
lab_data$ps_quintile <- cut(
    lab_data$ps,
    breaks = quantile(lab_data$ps, probs = 0:5 / 5),
    labels = paste("Q", 1:5, sep = ""),
    include.lowest = TRUE
)

# Check balance within quintiles
balance_by_quintile <- lab_data ▷
    group_by(ps_quintile, treatment) ▷
    summarise(
        n = n(),
        ps_mean = mean(ps),
        age_mean = mean(age),
        severity_mean = mean(baseline_severity),
        .groups = "drop"
    ) ▷
    arrange(ps_quintile, treatment)

kable(
    balance_by_quintile,
```
```

```
    digits = 2,
    caption = "Balance within propensity score quintiles"
)

# Estimate treatment effects within each quintile
quintile_effects <- lab_data |>
    group_by(ps_quintile) |>
    summarise(
        n_total = n(),
        n_treated = sum(treatment),
        n_control = sum(1 - treatment),
        outcome_treated = mean(outcome[treatment == 1]),
        outcome_control = mean(outcome[treatment == 0]),
        quintile_ate = outcome_treated - outcome_control,
        .groups = "drop"
    )

kable(
    quintile_effects,
    digits = 2,
    caption = "Treatment effects within propensity score
    quintiles"
)

# Overall stratified estimate (weighted by quintile
size)
stratified_ate <- weighted.mean(
    quintile_effects$quintile_ate,
    weights = quintile_effects$n_total
)

cat("Stratified ATE estimate:", round(stratified_ate,
2), "\n")

# Plot quintile effects
ggplot(quintile_effects, aes(x = ps_quintile, y =
quintile_ate)) +
```

```
    geom_col(fill = "steelblue", alpha = 0.7) +
    geom_hline(
        yintercept = stratified_ate,
        color = "red",
        linetype = "dashed"
    ) +
    labs(
        title = "Treatment Effects by Propensity Score
        Quintile",
        x = "Propensity Score Quintile",
        y = "Average Treatment Effect",
        caption = "Red line shows overall stratified
        estimate"
    ) +
    theme_minimal()
```

Task 5: Inverse Probability of Treatment Weighting (IPTW)

Your Task: Create IPTW weights and estimate the average treatment effect.

```
```{r}
#| label: task5-skeleton

# Create ATE weights
lab_data$iptw_weights <- ifelse(
  lab_data$treatment == 1,
  # YOUR CODE HERE: weight for treated units,
  # YOUR CODE HERE: weight for control units
)

# Check weight distribution
summary(lab_data$iptw_weights)

# Estimate weighted treatment effect
```
```

```
iptw_ate <- # YOUR CODE HERE: calculate weighted means
```
```

**Solution: Task 5**

````
```{r}
#| label: task5-solution
#| eval: false

# Create ATE weights
lab_data$iptw_weights <- ifelse(
    lab_data$treatment == 1,
    1 / lab_data$ps, # Weight for treated units
    1 / (1 - lab_data$ps) # Weight for control units
)

# Check weight distribution
weight_summary <- summary(lab_data$iptw_weights)
print(weight_summary)

# Identify extreme weights (potential issues)
extreme_weights <- sum(
    lab_data$iptw_weights >
quantile(lab_data$iptw_weights, 0.99)
)
cat(
    "Number of observations with extreme weights (>99th
    percentile):",
    extreme_weights,
    "\n"
)

# Visualize weight distribution
ggplot(lab_data, aes(x = iptw_weights)) +
    geom_histogram(bins = 50, fill = "lightblue", alpha
    = 0.7) +
````

```
    geom_vline(
        xintercept = median(lab_data$iptw_weights),
        color = "red",
        linetype = "dashed"
    ) +
    labs(
        title = "Distribution of IPTW Weights",
        x = "Weight",
        y = "Count",
        caption = "Red line shows median weight"
    ) +
    theme_minimal()

# Truncate extreme weights (optional stabilization)
lab_data$iptw_weights_trunc <- pmin(
    lab_data$iptw_weights,
    quantile(lab_data$iptw_weights, 0.99)
)

# Estimate weighted treatment effect
weighted_means <- lab_data |>
    group_by(treatment) |>
    summarise(
        weighted_outcome = weighted.mean(outcome, w =
        iptw_weights_trunc),
        .groups = "drop"
    )

iptw_ate <- diff(weighted_means$weighted_outcome)
cat("IPTW ATE estimate:", round(iptw_ate, 2), "\n")

# Check weighted balance
weighted_balance <- lab_data |>
    summarise(
        age_smd = abs(
            weighted.mean(
                age[treatment == 1],
```

```
                w = iptw_weights_trunc[treatment == 1]
            ) -
                weighted.mean(
                    age[treatment == 0],
                  w = iptw_weights_trunc[treatment == 0]
                )
        ) /
            sqrt(
                (wtd.var(
                    age[treatment == 1],
                    weights =
                    iptw_weights_trunc[treatment == 1]
                ) +
                    wtd.var(
                        age[treatment == 0],
                        weights =
                        iptw_weights_trunc[treatment ==
                        0]
                    )) /
                    2
            ),
        severity_smd = abs(
            weighted.mean(
                baseline_severity[treatment == 1],
                w = iptw_weights_trunc[treatment == 1]
            ) -
                weighted.mean(
                    baseline_severity[treatment == 0],
                  w = iptw_weights_trunc[treatment == 0]
                )
        ) /
            sqrt(
                (wtd.var(
                    baseline_severity[treatment == 1],
                    weights =
                    iptw_weights_trunc[treatment == 1]
                ) +
```

```
                    wtd.var(
                      baseline_severity[treatment == 0],
                        weights =
                        iptw_weights_trunc[treatment ==
                        0]
                    )) /
                    2
            )
    )

# Note: wtd.var function from Hmisc package - simplified
calculation here
cat("Weighted balance achieved (SMDs should be <0.1)\n")
```

Task 6: Compare All Methods

Your Task: Create a summary comparing all propensity score methods with the naive estimate.

```
```{r}
#| label: task6-solution

# Summary of all estimates
method_comparison <- tibble(
    Method = c(
        "Naive (Unadjusted)",
        "Propensity Score Matching",
        "Propensity Score Stratification",
        "IPTW",
        "True ATE"
    ),
    Estimate = c(naive_effect, match_ate,
    stratified_ate, iptw_ate, 5.0),
    `Sample Size` = c(
        nrow(lab_data),
```
```

```
        nrow(matched_data),
        nrow(lab_data),
        nrow(lab_data),
        NA
    ),
    Notes = c(
        "Biased due to confounding",
        "Reduced sample size",
        "Uses full sample",
        "Uses full sample",
        "Simulation truth"
    )
)

kable(
    method_comparison,
    digits = 2,
    caption = "Comparison of treatment effect estimates
    across methods"
)

# Visualize comparison
ggplot(method_comparison[1:4, ], aes(x = Method, y =
Estimate, fill = Method)) +
    geom_col(alpha = 0.7) +
    geom_hline(
        yintercept = 5,
        color = "red",
        linetype = "dashed",
        linewidth = 1
    ) +
    coord_flip() +
    labs(
        title = "Treatment Effect Estimates by Method",
        y = "Average Treatment Effect",
        caption = "Red line shows true ATE = 5"
    ) +
```

```
    theme_minimal() +
    theme(legend.position = "none")
```
```

#### Lab Reflection Questions

### Reflection Questions

1. **Which method gave the estimate closest to the true ATE of 5? Why might this be?**
2. **What are the trade-offs between matching (reduced sample size) and weighting (full sample but potential for extreme weights)?**
3. **How did the covariate balance change after applying propensity score methods? What does this tell us about confounding in the original data?**
4. **When might you choose stratification over the other methods?**
5. **What would you do if you found poor overlap in propensity scores between treatment groups?**

#### Retrieval Practice Quiz

### Quick Quiz: Test Your Understanding

**Question 1**: The propensity score is defined as: a) The probability of a good outcome given treatment b) The probability of treatment given the confounders c) The average treatment effect d) The correlation between treatment and outcome

**Answer**: b) The probability of treatment given the confounders

**Question 2**: Why does propensity score matching work? a) It eliminates all bias b) It creates balance on observed confounders c) It increases sample size d) It reduces measurement error
```

Answer: b) It creates balance on observed confounders - the balancing property ensures that within propensity score strata, treatment assignment is independent of the confounders.

Question 3: In IPTW, units with very low propensity scores receive: a) Low weights b) High weights c) Zero weights d) Average weights

Answer: b) High weights - Control units (A=0) with low propensity scores get weight 1/(1-ps), which is large when ps is small.

6 Common Pitfalls & Checks

Critical Assumptions and Their Violations

⚠ Pitfall 1: Ignoring Positivity Violations

Problem: Using propensity score methods when there's poor overlap between treatment groups can lead to biased estimates and inappropriate extrapolation.
Check: Always create overlap plots and examine the distribution of propensity scores in both treatment groups.

````
```{r}
#| label: positivity-check

# Check for positivity violations
ps_range <- range(lab_data$ps)
treated_range <-
range(lab_data$ps[lab_data$treatment == 1])
control_range <-
range(lab_data$ps[lab_data$treatment == 0])

cat("Overall PS range:", round(ps_range, 3), "\n")
cat("Treated PS range:", round(treated_range, 3),
"\n")
cat("Control PS range:", round(control_range, 3),
"\n")

# Flag observations in non-overlap regions
overlap_flag <- lab_data$ps >
max(min(treated_range), min(control_range)) &
    lab_data$ps < min(max(treated_range),
max(control_range))
cat("Observations in overlap region:",
sum(overlap_flag), "of", nrow(lab_data))
```
````

⚠ Pitfall 2: Extreme Weights in IPTW

Problem: Units with propensity scores near 0 or 1 receive extremely large weights, leading to unstable estimates dominated by a few observations.
Check: Examine weight distributions and consider truncation or trimming.

````
```{r}
#| label: extreme-weights-check

# Identify problematic weights
weight_percentiles <- quantile(
    lab_data$iptw_weights,
    probs = c(0.95, 0.99, 0.999)
)
print(weight_percentiles)

# Rule of thumb: weights >10 may be problematic
problematic_weights <- sum(lab_data$iptw_weights >
10)
cat("Observations with weights > 10:",
problematic_weights)

# Effective sample size
eff_n <- (sum(lab_data$iptw_weights))^2 /
sum(lab_data$iptw_weights^2)
cat(
    "Effective sample size:",
    round(eff_n, 1),
    "vs original n =",
    nrow(lab_data)
```
````

! Essential Diagnostics Checklist

Before Analysis: - [ ] Check for positivity/overlap between treatment groups - [ ] Examine propensity score model fit and specification - [ ] Assess baseline covariate imbalance
After Analysis: - [ ] Verify covariate balance achieved (SMD < 0.1 or 0.25) - [ ] Check for extreme weights (if using IPTW) - [ ] Ensure adequate sample sizes in each group/stratum - [ ] Sensitivity analysis for unmeasured confounding

Balance Assessment Guidelines

Standardized Mean Differences (SMD) Benchmarks

- **SMD < 0.1**: Excellent balance (preferred)
- **SMD < 0.25**: Acceptable balance
- **SMD ≥ 0.25**: Poor balance, adjustment needed

For binary variables, use the standardized difference in proportions: $SMD = \frac{|\hat{p}_{treat} - \hat{p}_{control}|}{\sqrt{\frac{\hat{p}_{treat}(1-\hat{p}_{treat}) + \hat{p}_{control}(1-\hat{p}_{control})}{2}}}$

Model Specification Considerations

! Propensity Score Model Building

Include: All confounders (variables that affect both treatment and outcome) **Consider**: - Interaction terms between confounders - Non-linear relationships (splines, polynomials) - Different link functions beyond logit
Don't: Include instrumental variables, colliders, or mediators
Goal: Predict treatment assignment, not the outcome

7 Advanced Teaching Activities & Interactivity

Think-Pair-Share Activity: Propensity Score Intuition

Activity 1: The Balancing Score Concept (15 minutes)

Individual Think (5 min): Students consider this scenario: "You want to study the effect of a new teaching method on student performance. You have data on 1,000 students, with information on their prior GPA, socioeconomic status, motivation level, and study time. Some students self-selected into the new teaching method."

Questions to ponder: 1. What confounding issues do you expect? 2. How might propensity scores help? 3. What would "balance" look like in this context?

Pair Discussion (5 min): Students discuss their thoughts with a partner.

Share (5 min): Pairs share insights with the class, instructor facilitates discussion of key concepts.

Problem-Based Learning Case

Case Study: Hospital Treatment Effectiveness

Scenario: You're analyzing whether a new cardiac procedure improves patient outcomes. You have data on 2,000 patients, but treatment assignment wasn't randomized - doctors chose treatments based on patient characteristics.

Available variables: - Age, gender, baseline cardiac function - Comorbidities, insurance type, hospital size - Treatment received, 30-day mortality outcome

Student Tasks: 1. Identify potential confounders and justify selections 2. Specify a propensity score model 3. Choose appropriate PS method and justify 4. Interpret results and assess limitations

Deliverable: 5-minute presentation of analysis plan and preliminary results

Interactive Simulation Exercise

Simulation: The Impact of Model Misspecification

Students use R to investigate what happens when: 1. Important confounders are omitted from the PS model 2. Non-linear relationships are ignored 3. Interaction effects are missing

Code Template:

```{r}

simulate_ps_bias <- function(
    include_interaction = TRUE,
    include_nonlinear = TRUE,
    n = 1000
) {
   # Generate data with interactions and non-linearities
    # Fit PS models with different specifications
    # Compare resulting treatment effect estimates
    # Return bias and MSE
}

# Students run simulations and discuss results
results <- expand.grid(
    interaction = c(TRUE, FALSE),
    nonlinear = c(TRUE, FALSE)
) |>
    mutate(bias = map2_dbl(interaction, nonlinear, ~
    simulate_ps_bias(.x, .y)))
```

Retrieval Practice with Spaced Intervals

Spaced Retrieval Questions

Week 5 (Current): 1. What is the balancing property of propensity scores? 2. When would you choose matching over weighting?

Week 7 Review (Interleaved with new content): 3. How do propensity scores relate to exchangeability? 4. What diagnostics would you use for IPTW?

Week 10 Review (Before exam): 5. Compare propensity scores to direct covariate adjustment 6. Design a study using propensity score stratification

Peer Instruction Sequence

Concept Test: Propensity Score Applications

Question: You have estimated propensity scores for a study. The treated group has PS ranging from 0.1 to 0.9, while the control group ranges from 0.05 to 0.7. Which approach would be MOST appropriate?

A) Use all observations with IPTW
B) Match on propensity scores using all observations
C) Restrict analysis to PS range 0.1-0.7 before matching
D) Stratify into quintiles using all observations

Teaching Sequence: 1. Students vote individually (polling) 2. Discuss with neighbors if <70% correct 3. Re-vote after discussion 4. Instructor explains correct answer (C)

Self-Explanation Prompts

Explain Your Reasoning

After completing each lab task, students write brief explanations:

1. **After matching**: "Explain in your own words why the matched sample is more appropriate for causal inference than the original sample."
2. **After IPTW**: "Describe what the 'pseudo-population' created by IPTW looks like and why treatment effects estimated in this population are interpretable as causal effects."
3. **After balance checking**: "If you found poor balance on one covariate after propensity score adjustment, what would you do and why?"

8 Appendix: Advanced Theory & Derivations

Theoretical Foundation: The Fundamental Theorem

! Rosenbaum-Rubin Theorem (1983)

If assignment to treatment is strongly ignorable given covariates $\mathbf{X}$: $(\mathbf{Y}^0, \mathbf{Y}^1) \perp A|\mathbf{X}$
Then assignment to treatment is also strongly ignorable given the propensity score $e(\mathbf{X})$: $(\mathbf{Y}^0, \mathbf{Y}^1) \perp A|e(\mathbf{X})$
Proof Sketch: The propensity score is a balancing score, meaning: $\mathbf{X} \perp A|e(\mathbf{X})$
Combined with the assumption $(\mathbf{Y}^0, \mathbf{Y}^1) \perp A|\mathbf{X}$, this implies the desired result through the chain of conditional independence relationships.

Mathematical Properties of Propensity Score Methods

Asymptotic Properties of Matching Estimators

For 1:1 nearest neighbor propensity score matching, the bias of the treatment effect estimator is:

$\text{Bias}(\hat{\tau}_{match}) \approx \frac{1}{2}\mu_0''(\bar{e})\sigma_e^2 + O_p(n^{-1/2})$

where $\mu_0(e)$ is the conditional expectation of Y^0 given the propensity score, and σ_e^2 is the variance of propensity score differences within matched pairs.

Numerical Illustration: Bias of Matching

Consider a simple case where: - $\mu_0(e) = 10 + 5e + 2e^2$ (quadratic relationship) - $\mu_0''(e) = 4$ (second derivative) - $\sigma_e^2 = 0.01$ (small matching discrepancies)

Then: Bias $\approx \frac{1}{2}(4)(0.01) = 0.02$

The bias is small when matching is tight (σ_e^2 small) and the outcome-propensity score relationship is approximately linear.

Efficiency of IPTW Estimators

The asymptotic variance of the IPTW estimator for the ATE is:

$\text{Var}(\hat{\tau}_{IPTW}) = E\left[\frac{\text{Var}(Y^1|X)}{e(X)} + \frac{\text{Var}(Y^0|X)}{1-e(X)} + (Y^1 - Y^0 - \tau)^2\right]$

This shows that IPTW is most efficient when: 1. Propensity scores are close to 0.5 (balanced design) 2. Conditional outcome variances are small 3. Treatment effect heterogeneity is minimal

Numerical Example: IPTW Efficiency

Assume $\text{Var}(Y^1|X) = \text{Var}(Y^0|X) = 25$ and $\tau = 5$ (constant treatment effect).

For different propensity scores: - $e(X) = 0.1$: Var $\approx \frac{25}{0.1} + \frac{25}{0.9} = 250 + 27.8 = 277.8$ - $e(X) = 0.5$: Var $\approx \frac{25}{0.5} + \frac{25}{0.5} = 50 + 50 = 100$ - $e(X) = 0.9$: Var $\approx \frac{25}{0.9} + \frac{25}{0.1} = 27.8 + 250 = 277.8$

The variance is minimized when propensity scores are balanced around 0.5.

Advanced Diagnostics: The Overlap Assumption

Crump Et Al. (2009) Trimming Rule

To optimize bias-variance tradeoffs in IPTW, Crump et al. (2009) propose trimming observations with propensity scores outside the interval:

$[\alpha, 1-\alpha]$ where $\alpha = \frac{0.1}{\sqrt{k/n}}$

where k is the number of covariates and n is the sample size.

Numerical Application: Crump Trimming

For our lab example with $k = 6$ covariates and $n = 800$:

$\alpha = \frac{0.1}{\sqrt{6/800}} = \frac{0.1}{\sqrt{0.0075}} = \frac{0.1}{0.087} = 1.15$

Since $\alpha > 1$, no trimming is recommended (the formula suggests we need more data relative to the number of covariates for reliable trimming).

Doubly Robust Estimation

The AIPW Estimator

The Augmented Inverse Probability Weighted (AIPW) estimator combines outcome modeling with propensity score weighting:

$\hat{\tau}_{AIPW} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{A_i(Y_i-\hat{\mu}_1(X_i))}{\hat{e}(X_i)} - \frac{(1-A_i)(Y_i-\hat{\mu}_0(X_i))}{1-\hat{e}(X_i)} + \hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)\right]$

where $\hat{\mu}_a(X)$ are fitted outcome models for each treatment level.

Double Robustness Property: $\hat{\tau}_{AIPW}$ is consistent if either the propensity score model OR the outcome models are correctly specified (but not necessarily both).

Implementation Example

```
```{r}

# Fit outcome models
mu1_model <- lm(outcome ~ age + baseline_severity +
comorbidities,
              data = lab_data[lab_data$treatment == 1,])
mu0_model <- lm(outcome ~ age + baseline_severity +
comorbidities,
              data = lab_data[lab_data$treatment == 0,])

# Predict outcomes for all units
mu1_hat <- predict(mu1_model, newdata = lab_data)
mu0_hat <- predict(mu0_model, newdata = lab_data)

# Calculate AIPW estimate
aipw_components <- with(lab_data, {
  term1 <- treatment * (outcome - mu1_hat) / ps
```
```

```
  term2 <- (1 - treatment) * (outcome - mu0_hat) / (1 -
ps)
  term3 <- mu1_hat - mu0_hat
  term1 - term2 + term3
})

tau_aipw <- mean(aipw_components)
```

References

Crump, R. K., Hotz, V. J., Imbens, G. W., & Mitnik, O. A. (2009). Dealing with limited overlap in estimation of average treatment effects. *Biometrika*, 96(1), 187-199.

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